GridfinityGeneratorPlus

Version: 1.4.0.0

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A Fusion add-in designed to make generating, previewing, and revisiting Gridfinity parts fast, intuitive, and future-proof with creation settings that aren't lost to time.

Introduction: The "Why" and "What"

The original GridfinityGenerator add-in is a great parametric bin/baseplate generator, but it works through two separate command dialogs that don't share settings, offer no preview before committing, and forget everything about how a part was built the moment you click OK — so revisiting a design months later, or handing the file to someone else, means starting from scratch. GridfinityGeneratorPlus keeps the original's proven geometry engine and rebuilds the interface and workflow around it.

GridfinityGeneratorPlus is a Fusion add-in designed to make generating, previewing, and revisiting Gridfinity parts fast and durable.

- **Feature 1: One persistent palette, not two dialogs.** Bin, Baseplate, Grid Optimizer, and Theme all live in a single dockable palette with shared "Common settings," live Update Preview /

Generate / Clear Preview, and named saved configurations per tab.

- **Feature 2: Settings travel with the file.** Every generated part remembers the exact parameters that built it, stored directly on the component itself. Reopen a design months later, or share the `.f3d` with someone else running this add-in, and click **Edit Active Component** to pick up right where you left off.
- **Feature 3: Grid Optimizer.** Building a custom (non-42mm) grid to fit a specific drawer or cabinet? Enter your real measurements and let the optimizer recommend a pitch that minimizes wasted space, instead of guessing.
- **Feature 4: Stackable baseplates.** A fourth baseplate type, made symmetric top-to-bottom so it can be printed and physically stacked with no overhang, with configurable interface layers between plates.
- **Feature 5: Themes tab.** Customize the palette's appearance — built-in themes, automatic OS dark-mode following, and import/export of your own custom themes as either a single `.theme.json` file or a multi-theme `style.css` bundle.
- **Feature 6: Half-unit grid support & DXF export for laser cutting.** Add a half-unit-wide strip to any edge (Left/Right/Front/Back) of a bin or baseplate — where two checked edges meet at a corner, that cell automatically becomes quarter-size, no extra configuration needed. Baseplates can also generate a DXF-ready sketch for laser-cut fabrication.

✨ What's New in v1.4.0

- **Fleet UI standardization, completed.** This add-in's palette now matches the structural standard shared across Ed's Fusion add-in fleet: the "Theme" tab is renamed **Themes** (plural), and **Common Settings** is now a collapsible section like every other panel in the palette, instead of an always-expanded box.
- **CSS-bundle theme import/export.** Alongside the existing single-theme `.theme.json` import/export, the Themes tab can now import/export a multi-theme `style.css` bundle (one `:root[data-theme="Name"]` block per custom theme) — useful for backing up or sharing several custom themes at once.
- **Factory Reset for Themes.** A new Factory Reset button on the Themes tab permanently clears every imported theme and resets font settings back to default (with a confirmation prompt first, since it can't be undone).
- **Accessibility pass.** Every button's text color now goes through a themed `--btn-primary-text` / `--btn-danger-text` variable instead of a hardcoded white, checked against WCAG AA contrast for all four built-in themes (Midnight and Sandstone's bright accent buttons needed dark text, not white, to stay readable). Keyboard (Tab) focus is now visibly outlined on

every button, input, dropdown, and collapsible section header, without adding an outline on mouse clicks. The confirmation-modal backdrop is also now themed instead of a fixed black tint.

✨ What's New in v1.3.0

- **New palette header.** The palette now has a header bar at the top with the add-in title, version, and the theme selector (moved here from the Theme tab, which still holds font family/size, import/export, and Remove Selected Theme).
- **Tab bar visual fix.** Corrected inconsistent CSS that was giving every tab button a boxed border instead of the intended flat underline style.

✨ What's New in v1.2.1

- **Fixed a crash generating Shelled bins.** Shelled-type bins could fail to generate with a `Shell1 / Compute Failed // ASM_REM_NO_SOLUTION` error, caused by a bin-foot/wall alignment change shipped in v1.2.0 (see below). Reverted — Shelled bins generate correctly again.
- **Fixed off-center baseplate cutout openings.** Baseplate grid openings had `xyClearance` more slack on the back/right of each cell than the front/left, so bin feet sat visibly off-center in the opening. Now centered evenly on all four sides.

✨ What's New in v1.2.0

- **Half-unit edge support for bins and baseplates.** New Left/Right/Front/Back checkboxes on both tabs add a half-unit-wide strip to that edge; where two adjacent checked edges meet at a corner, that cell becomes quarter-size automatically. Core width/length can now be 0 when half-edges supply the rest of that axis — build a bin or baseplate entirely out of half- and quarter-size cells if you want to. Magnet/screw cutouts only ever apply to true full-size cells; a Skeletonized baseplate's decorative bottom groove is likewise full-cells-only, with half/quarter cells staying solid.
- **DXF-ready sketch for laser-cut baseplates.** New "Generate DXF-ready sketch" option on the Baseplate tab adds a sketch of the plate's mid-height cross-section that you can export to DXF/SVG by hand from Fusion, for laser-cutting baseplates out of thin sheet material.
- ~~**Bin foot alignment fix.** Bin stacking feet now center correctly within each baseplate grid opening — previously they sat `xyClearance` off-center, most noticeable on precise/tight-clearance prints.~~ *(Superseded by v1.2.1 — this fix was incomplete and caused a Shelled-bin crash; see above.)*

✨ What's New in v1.1.2

- **Fixed tripled bodies in stacked baseplates:** generating a Stackable baseplate with Stack Count > 1 silently produced 3x the expected baseplate/interface-layer bodies, all coincident in the Fusion viewport (so it looked correct there) but appearing as extra duplicate solids once exported/sliced. Root cause was a shared pattern-feature helper that only configured its first direction, leaving Fusion's own default second-direction quantity (3) active. If you've generated any Stackable baseplates with Stack Count > 1, regenerate them to pick up the fix.

✨ What's New in v1.1.1

- **Fixed a notch in the bin base's bottom edge:** multi-cell bins previously showed a small visible notch at every interior grid seam along the outer bottom edge. The base is now built at its final, clearance-adjusted size from the very first sketch instead of being reshaped by a separate step afterward, so the outer edge comes out clean — you'll now see a small (expected, spec-accurate) gap between adjacent unit cells' feet instead.
- **Dependent checkbox fixes:** unchecking **With lip** now also unchecks and disables **With lip notches** (re-enabling it when you check lip back on); unchecking **Add magnet cutouts** does the same for **Add magnet cutout tabs** — previously these could be left checked while their parent option was off.

For the full development history and the reasoning behind each design decision, see `docs/Dev_Notes.md`.

Installation

> **Tip:** For a stable, versioned copy, download the latest packaged release from the **Releases** link in the right-hand sidebar of this repo's GitHub page instead of using the green **Code** button above — the Code button always pulls the current `main` branch, which may include in-progress changes. Or, simply click this link to the Releases page

Manual Installation Options

This add-in requires a quick manual installation. You can choose to install it in Fusion's default directory or a custom folder of your choice.

Option 1: Install in the Default Fusion Directory

1. **Download:** Download the source code as a ZIP file and extract the `GridfinityGeneratorPlus-main` folder. Rename the folder to `GridfinityGeneratorPlus` (remove the `-main` suffix) — Fusion requires the folder name to match the add-in name exactly, so it won't run correctly if you skip this step.
   Download the zip file using the green `Code` button above or simply click this link:
   [GridfinityGeneratorPlus Main Branch]
2. **Move the Folder:** Move the entire `GridfinityGeneratorPlus` folder into your native Fusion Add-Ins directory:
   - **Windows:** `%appdata%\Autodesk\Autodesk Fusion 360\API\Addins`
   - **Mac:** `~/Library/Application Support/Autodesk/Autodesk Fusion 360/API/Addins`
3. **Open Fusion:** Press `Shift + S` to open the **Scripts and Add-Ins** dialog.
4. **Run the Add-in:** Make sure the **Add-ins** filter checkbox is checked. You should see **GridfinityGeneratorPlus** in the list of add-ins. You may want to check the 'Run on startup' option so it automatically runs when Fusion starts. Click the **Run** icon to execute the add-in.

Option 2: Install in a Custom Directory

1. **Download:** Download the source code as a ZIP file and extract the `GridfinityGeneratorPlus` folder. Rename the folder to `GridfinityGeneratorPlus`.
2. **Organize:** Create a dedicated folder on your computer for your Fusion tools (e.g., `Documents\Fusion_Tools`) and move the `GridfinityGeneratorPlus` folder inside it (remove the `-main` suffix) — Fusion requires the folder name to match the add-in name exactly, so it won't run correctly if you skip this step.
3. **Open Fusion:** Press `Shift + S` to open the **Scripts and Add-Ins** dialog.
4. **Add the Add-in:** Click the grey **"+"** icon next to the search box at the top of the dialog and select **Script or add-in from device**.
5. **Locate:** Navigate to your custom folder, select the `GridfinityGeneratorPlus` folder, and click **Select Folder**.
6. **Run the Add-in:** Make sure the **Add-ins** filter checkbox is checked. You should see **GridfinityGeneratorPlus** in the list of add-ins. You may want to check the 'Run on startup' option so it automatically runs when Fusion starts. Click the **Run** icon to execute the add-in.

Using GridfinityGeneratorPlus

The Gridfinity Generator Palette

In the **DESIGN** workspace, find the **SOLID** toolbar panel's **CREATE** dropdown and select **"GridfinityGeneratorPlus"** to open the palette. It stays docked and open across your session — set your Common settings once, then switch between tabs as needed.

- **Common settings panel:** base grid unit, XY clearance, and magnet cutout size, shared by both Bin and Baseplate so they always stay compatible with each other.
- **Update Preview / Generate / Clear Preview:** preview a part in the viewport before committing to it, make it permanent when you're happy, or discard the preview entirely.
- **Configurations:** save, load, and delete named presets per tab (Bin, Baseplate, Grid Optimizer), so a frequently-used setup is always one click away.

Edit Active Component

Activate a previously generated Gridfinity part in the Fusion browser by hovering over the component name and clicking the radio button that appears to the righ of the name, then click **Edit Active Component** at the top of the palette — it reloads the exact settings used to build that part.

- **Tip:** this requires the design to be in **Hybrid** design mode (Document Settings → Design → Hybrid). If you click **Generate** in a Part or Assembly design, the palette will prompt you to switch modes — your entered values aren't lost, so just switch and click Generate again.

For the complete usage guide, including every tab, field, and workflow in detail, see GGPlus_Help.md.

Tech Stack

For the fellow coders and makers out there, here is how GridfinityGeneratorPlus was built:

- **Language:** Python (Fusion API)
- **Interface:** HTML/CSS/JavaScript (Palette)
- **Data Storage:** Each generated part's settings are stored as a Fusion component attribute directly tied to the component, so they travel with the `.f3d` file itself. Add-in preferences (window geometry, active theme), saved configurations, and imported themes are stored locally as JSON alongside the add-in.

Acknowledgements & Credits

- **Gridfinity by Zack Freedman**
- **Original Work:** GridfinityGeneratorPlus is a fork of **GridfinityGenerator by Lev Mishin**. The core parametric geometry — the base/foot profile, bin body construction, and baseplate generation — originates from his original work, without which this fork wouldn't exist.
- **Developer:** Ed Johnson (Making With An EdJ)
- **AI Assistance:** Developed with coding assistance from Claude (Anthropic).
- **Lucy (The Cavachon Puppy):**
  Chief Wellness Officer & Director of Mandatory Breaks
  - Thank you for ensuring I maintained healthy circulation by interrupting my deep coding sessions with urgent requests for play.

❤️ Support the Maker (and Lucy!)

I develop these tools to improve my own workflows and love sharing them with the community. If you find GridfinityGeneratorPlus useful and want to say thanks, feel free to **buy Lucy a dog treat on Ko-fi**!

Happy Making!
— EdJ